

Averaged Intercept Envelopes for Random Polynomial Root Hitting

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1 Theoretical Setup

1.1 Problem Setting

Fix an integer $d \geq 1$, a radius $R > 0$, and a compact one-dimensional parameter domain $\Theta \subseteq \mathbb{R}$. Let $\mathcal{D}$ be a nonempty class of Borel probability laws on $[-R, R]^d$. Write

$$\alpha = (\alpha_0, \beta), \quad \beta = (\alpha_1, \dots, \alpha_{d-1}), \quad \phi_\alpha(\theta) = \theta^d + \sum_{k=0}^{d-1} \alpha_k \theta^k.$$

For $d = 1$, β is the empty tuple and its marginal is the point mass on the one-point space. For any Borel probability law ν on $[-R, R]^d$, denote its β -marginal by π_ν and write $\Pr_\nu$ for probability when $\alpha \sim \nu$. No simplicity, root-separation, or transversality condition is imposed on the zeros of ϕ_α , and no learner or online algorithm is fixed.

Let $\mathcal{I}(\Theta)$ be the collection of all intervals $I \subseteq \Theta$ with positive Lebesgue length $|I| > 0$, allowing open, closed, and half-open endpoints, and put

$$m_I = \frac{\inf I + \sup I}{2}, \quad Z_\alpha = \{\theta \in \Theta : \phi_\alpha(\theta) = 0\}.$$

For every nonempty class $\mathcal{A}$ of cube-supported coefficient laws, define

$$C_{\mathcal{A}} := \sup_{\nu \in \mathcal{A}} \sup_{I \in \mathcal{I}(\Theta)} \frac{\Pr_\nu(Z_\alpha \cap I \neq \emptyset)}{|I|}.$$

In particular, the primary target uses $\mathcal{A} = \mathcal{D}$.

Put

$$B := \max \left\{ 1, \sup_{\theta \in \Theta} |\theta| \right\}, \quad L_{d,R,\Theta} := \sup_{\substack{\alpha \in [-R,R]^d \\ \theta \in \Theta}} |\partial_\theta \phi_\alpha(\theta)|.$$

Since

$$\partial_\theta \phi_\alpha(\theta) = d\theta^{d-1} + \sum_{k=1}^{d-1} k\alpha_k \theta^{k-1},$$

the deterministic envelope is finite and satisfies

$$L_{d,R,\Theta} \leq dB^{d-1} + R \sum_{k=1}^{d-1} kB^{k-1} < \infty,$$

where an empty sum is zero.

Whenever a cube-supported coefficient law ν has a regular conditional density $f_\nu(a_0 \mid \beta)$ of α_0 given β , extend that density by zero outside $[-R, R]$ and define

$$K_\nu(\beta) := \operatorname{ess\,sup}_{a_0 \in \mathbb{R}} f_\nu(a_0 \mid \beta).$$

For any class $\mathcal{A}$ of such laws, define its averaged envelope by

$$\bar{\kappa}_{\mathcal{A}} := \sup_{\nu \in \mathcal{A}} \int K_\nu(\beta) \pi_\nu(d\beta).$$

For a constructive specialization, let $\mathcal{D}_{\text{RI}}$ be a nonempty class of random-intercept laws. For each $\mu \in \mathcal{D}_{\text{RI}}$, the vector $\beta \sim \pi_\mu$ has an arbitrary law on $[-R, R]^{d-1}$, $U_\mu \sim \text{Unif}[-1, 1]$ is independent of β , and measurable functions G_μ, ρ_μ generate

$$\alpha_0 = G_\mu(\beta) + \rho_\mu(\beta)U_\mu.$$

A fixed-width member is one for which $\rho_\mu(\beta) = r_\mu$ for π_μ -almost every β , for some constant $r_\mu > 0$. The notation $\mathbf{1}\{E\}$ denotes the indicator of an event E .

For the explicit heteroscedastic witness, whenever $d \geq 3$ and $q \in (0, 1)$, let

$$Z \sim \text{Unif}[-R/2, R/2], \quad U \sim \text{Unif}[-1, 1]$$

be independent, and define

$$\rho_q(z) := \frac{R}{2} \left(\frac{2|z|}{R} \right)^q, \quad \beta_q(z) := (z, z^2/R, 0, \dots, 0), \quad \alpha^{(q)} := (\rho_q(Z)U, \beta_q(Z)).$$

Denote the law of $\alpha^{(q)}$ by μ_q .

1.2 Assumptions

Assumption 1 (Averaged conditional intercept-density envelope). *For every $\mu \in \mathcal{D}$, a regular conditional law of α_0 given β admits a jointly measurable Lebesgue density $f_\mu(a_0 \mid \beta)$ for π_μ -almost every β ; the corresponding K_μ is π_μ -measurable; and $\bar{\kappa}_{\mathcal{D}} < \infty$. The marginal π_μ is otherwise unrestricted: it may be discrete or singular and its coordinates may be arbitrarily dependent. In particular, this assumption imposes no finite bound on $\operatorname{ess\,sup}_{\beta \sim \pi_\mu} K_\mu(\beta)$, either for an individual μ or uniformly over $\mathcal{D}$.*

Assumption 2 (Random-intercept construction). *For every $\mu \in \mathcal{D}_{\text{RI}}$, the functions in the random-intercept representation satisfy $\rho_\mu(\beta) > 0$ for π_μ -almost every β and*

$$|G_\mu(\beta)| + \rho_\mu(\beta) \leq R \quad \text{for } \pi_\mu\text{-almost every } \beta, \quad M_{\text{RI}} := \sup_{\mu \in \mathcal{D}_{\text{RI}}} \int \frac{1}{\rho_\mu(\beta)} \pi_\mu(d\beta) < \infty.$$

This is a sufficient construction used to verify Assumption 1, not an additional requirement on a general class $\mathcal{D}$. No density, absolute continuity, or coordinate-independence condition is imposed on π_μ .

1.3 Formalized Goal

The goal is a one-way coefficient-side sufficient theorem. Under Assumption 1, simultaneously for every $\mu \in \mathcal{D}$ and every $I \in \mathcal{I}(\Theta)$, it should establish

$$\Pr_{\mu}(Z_{\alpha} \cap I \neq \emptyset) \leq \bar{\kappa}_{\mathcal{D}} L_{d,R,\Theta} |I|, \quad C_{\mathcal{D}} \leq \bar{\kappa}_{\mathcal{D}} L_{d,R,\Theta} < \infty.$$

Under Assumption 2, it should derive the exact conditional density and cap

$$f_{\mu}(a_0 | \beta) = \frac{\mathbf{1}\{|a_0 - G_{\mu}(\beta)| \leq \rho_{\mu}(\beta)\}}{2\rho_{\mu}(\beta)}, \quad K_{\mu}(\beta) = \frac{1}{2\rho_{\mu}(\beta)}, \quad \bar{\kappa}_{\mathcal{D}_{\text{RI}}} \leq \frac{1}{2} M_{\text{RI}},$$

and the corresponding all-law, all-positive-length-interval conclusion, including fixed-width members with arbitrary possibly singular or dependent higher-coordinate marginals.

For every $d \geq 3$, $R > 0$, and $q \in (0, 1)$, the same theorem should show that μ_q is supported on $[-R, R]^d$, that $\rho_q(Z) > 0$ almost surely, and that

$$\text{ess sup}_{\beta \sim \pi_{\mu_q}} K_{\mu_q}(\beta) = \infty, \quad \int K_{\mu_q}(\beta) \pi_{\mu_q}(d\beta) = \frac{1}{R(1-q)}, \quad C_{\{\mu_q\}} \leq \frac{L_{d,R,\Theta}}{R(1-q)}.$$

It should also identify the exact support

$$\{(a_0, z, z^2/R, 0, \dots, 0) : |z| \leq R/2, |a_0| \leq \rho_q(z)\}$$

and prove, in the precisely limited full-column-rank affine-latent baseline with an absolutely continuous bounded-density latent law, that the curved (α_1, α_2) projection prevents affine-latent representability. The result is sufficient only: it asserts neither necessity nor a converse, and makes no uniform polynomial-in- (d, R) claim for an indexed family.

2 Preliminaries

For a Borel map F and probability law P , $F_{\#}P$ denotes the pushforward law. We write $\text{supp}(P)$ for topological support, $\text{aff}(A)$ for affine hull, $\mathcal{L}^k$ for k -dimensional Lebesgue measure, and $\mathcal{H}_L^k$ for the induced k -dimensional Hausdorff volume on a k -dimensional affine space L . These notions are used only in the structural witness clause of the main theorem and its proof.

All quantitative bounds below are literal finite-dimensional, static inequalities. There is no temporal horizon, limiting horizon, confidence parameter, or hidden multiplicative constant. Polynomial evaluations and derivatives use ordinary absolute value, coefficient and latent spaces use Euclidean norm when a norm is needed, interval size is Lebesgue length, and $C_{\mathcal{A}}$ is normalized by that same length.

3 Main Theorem

Theorem 3.1 (Averaged-intercept root hitting and a heteroscedastic witness). *Fix the basic setting above.*

General averaged-envelope conclusion. Under Assumption 1, simultaneously for every $\mu \in \mathcal{D}$ and every $I \in \mathcal{I}(\Theta)$,

$$\Pr_{\mu}(Z_{\alpha} \cap I \neq \emptyset) \leq L_{d,R,\Theta} |I| \int K_{\mu}(\beta) \pi_{\mu}(d\beta) \leq \bar{\kappa}_{\mathcal{D}} L_{d,R,\Theta} |I|,$$

and therefore

$$C_{\mathcal{D}} \leq \bar{\kappa}_{\mathcal{D}} L_{d,R,\Theta} < \infty, \quad L_{d,R,\Theta} \leq dB^{d-1} + R \sum_{k=1}^{d-1} kB^{k-1}.$$

The sum is zero and $L_{1,R,\Theta} = 1$ when $d = 1$.

Random-intercept conclusion. Separately, under Assumption 2, every $\mu \in \mathcal{D}_{\text{RI}}$ is supported on $[-R, R]^d$ and admits a jointly measurable conditional-density version for which, for π_{μ} -almost every β ,

$$f_{\mu}(a_0 \mid \beta) = \frac{\mathbf{1}\{|a_0 - G_{\mu}(\beta)| \leq \rho_{\mu}(\beta)\}}{2\rho_{\mu}(\beta)}, \quad K_{\mu}(\beta) = \frac{1}{2\rho_{\mu}(\beta)}.$$

Moreover,

$$\int K_{\mu} d\pi_{\mu} = \frac{1}{2} \int \rho_{\mu}^{-1} d\pi_{\mu}, \quad \bar{\kappa}_{\mathcal{D}_{\text{RI}}} = \frac{1}{2} M_{\text{RI}},$$

and, for every $I \in \mathcal{I}(\Theta)$,

$$\Pr_{\mu}(Z_{\alpha} \cap I \neq \emptyset) \leq \frac{L_{d,R,\Theta}|I|}{2} \int \rho_{\mu}^{-1} d\pi_{\mu}, \quad C_{\mathcal{D}_{\text{RI}}} \leq \frac{1}{2} L_{d,R,\Theta} M_{\text{RI}}.$$

If $\rho_{\mu} = r_{\mu} > 0$ almost surely, then $K_{\mu} = 1/(2r_{\mu})$ almost surely and

$$\Pr_{\mu}(Z_{\alpha} \cap I \neq \emptyset) \leq \frac{L_{d,R,\Theta}}{2r_{\mu}} |I|.$$

More generally, every nonempty class $\mathcal{F}$ of such fixed-width members satisfying the support condition and $\sup_{\mu \in \mathcal{F}} r_{\mu}^{-1} < \infty$ obeys

$$\bar{\kappa}_{\mathcal{F}} = \frac{1}{2} \sup_{\mu \in \mathcal{F}} r_{\mu}^{-1}, \quad C_{\mathcal{F}} \leq \frac{L_{d,R,\Theta}}{2} \sup_{\mu \in \mathcal{F}} r_{\mu}^{-1} < \infty.$$

No density or coordinate-independence condition on the higher-coordinate marginal is needed.

Explicit heteroscedastic witness. For every $d \geq 3$, $R > 0$, and $q \in (0, 1)$, the law μ_q satisfies

$$\text{supp}(\mu_q) = S_q := \{(a_0, z, z^2/R, 0, \dots, 0) : |z| \leq R/2, |a_0| \leq \rho_q(z)\} \subseteq [-R, R]^d$$

and $\Pr(\rho_q(Z) > 0) = 1$. One jointly measurable conditional-density version satisfies, for every $a_0 \in \mathbb{R}$ and $0 < |z| \leq R/2$,

$$f_{\mu_q}(a_0 \mid \beta_q(z)) = \frac{\mathbf{1}\{|a_0| \leq \rho_q(z)\}}{2\rho_q(z)}, \quad K_{\mu_q}(\beta_q(z)) = \frac{1}{R(2|z|/R)^q}.$$

The null conditioning fiber $z = 0$ is assigned a genuine fallback density, and

$$\text{ess sup}_{\beta \sim \pi_{\mu_q}} K_{\mu_q}(\beta) = \infty, \quad \int K_{\mu_q} d\pi_{\mu_q} = \frac{1}{R(1-q)}.$$

Consequently, for every $I \in \mathcal{I}(\Theta)$,

$$\Pr_{\mu_q}(Z_{\alpha} \cap I \neq \emptyset) \leq \frac{L_{d,R,\Theta}|I|}{R(1-q)}, \quad C_{\{\mu_q\}} \leq \frac{L_{d,R,\Theta}}{R(1-q)}.$$

The affine hull of S_q has dimension three and $\mathcal{H}_{\text{aff}(S_q)}^3(S_q) = 0$. More precisely, define the fixed-monic descending-coordinate embedding

$$\iota_d(a_0, a_1, \dots, a_{d-1}) = (1, a_{d-1}, \dots, a_0).$$

There do not exist an integer $k \geq 1$, a bounded-support random vector $X \in \mathbb{R}^k$ with a bounded Lebesgue density, and an affine map $x \mapsto b + Ax$ with $A \in \mathbb{R}^{(d+1) \times k}$ and $\text{rank}(A) = k$ such that $b + AX$ has law $(\iota_d)_{\#} \mu_q$. This exclusion does not assert anything about rank-deficient maps paired with singular latent laws.

Every displayed factor is exact. The exposed quantities are $d, R, \Theta, B, L_{d,R,\Theta}, |I|, \bar{\kappa}_{\mathcal{D}}, M_{\text{RI}}, \rho_{\mu}, r_{\mu}$, and, for the witness, q ; there are no hidden constants. The setting fixes d, R, Θ and the relevant class before laws and intervals vary, and the witness fixes d, R, q, Θ before its singleton calculation. The probability inequalities are deterministic statements about exact probabilities, the conditional identities hold on conditioning-marginal full-measure sets, and the width statement is almost sure. There is no temporal horizon. The norm and measure conventions are those stated in the setup and preliminaries.

These are one-way sufficient conclusions. They assert no necessity, converse, indexed uniform polynomial dependence on (d, R) , root simplicity, root separation, or transversality, and provide neither a learner nor an online guarantee.

Corollary 3.1 (Bounded-joint-density baseline). *Fix the basic setting and let $\mathcal{A}$ be a nonempty class of laws on $[-R, R]^d$. Suppose every $\nu \in \mathcal{A}$ has a Borel joint density $h_{\nu}(a_0, \beta)$, extended by zero outside the cube, satisfying $0 \leq h_{\nu} \leq \kappa_{\text{joint}} < \infty$ almost everywhere for one common constant κ_{joint} . Then*

$$\bar{\kappa}_{\mathcal{A}} \leq \kappa_{\text{joint}} (2R)^{d-1},$$

and, for every $\nu \in \mathcal{A}$ and $I \in \mathcal{I}(\Theta)$,

$$\Pr_{\nu}(Z_{\alpha} \cap I \neq \emptyset) \leq \kappa_{\text{joint}} (2R)^{d-1} L_{d,R,\Theta} |I|, \quad C_{\mathcal{A}} \leq \kappa_{\text{joint}} (2R)^{d-1} L_{d,R,\Theta}.$$

If $B = 1$, this becomes the exact monic baseline scale

$$C_{\mathcal{A}} \leq \kappa_{\text{joint}} (2R)^{d-1} \left(d + \frac{Rd(d-1)}{2} \right).$$

For $d = 1$, $(2R)^0 = 1$ and the bracket equals one. The displayed variables $\kappa_{\text{joint}}, d, R, B, L_{d,R,\Theta}$ and $|I|$ are all exposed, and there are no hidden constants. The statement is a static deterministic probability bound, uniform over the fixed class and all positive-length intervals, with the same norm and horizon conventions as Theorem 3.1.

4 Proof Sketch

A root in an interval forces the polynomial's value at the interval midpoint into a deterministic slab. Lemma A.1 gives the uniform derivative envelope, Proposition A.1 handles all endpoint conventions without selecting a root, and Proposition A.2 turns the root event into an intercept interval of exact length $L_{d,R,\Theta} |I|$. Conditional disintegration and integration of the fiber cap then yield Proposition A.3.

For random intercepts, Lemma A.4 constructs a measurable regular conditional kernel, including a legal fallback on null fibers. Propositions A.4 and A.5 derive the exact reciprocal-width budget before invoking the general result; Proposition A.6 records the fixed-width specialization.

The explicit witness is a continuous pushforward of a full-support latent rectangle. Propositions A.7 and A.8 identify its exact support and prove cube membership and almost-sure positive width. Lemma A.5 recovers the latent coordinate from the conditioning vector, and Proposition A.9 gives the exact cap. Positive-mass punctured neighborhoods prove its essential unboundedness, while a direct t^{-q} integral proves its finite mean; these are Lemmas A.6 and A.7.

For the structural comparison, the support has a three-dimensional affine hull but zero induced three-volume. The fixed-monic embedding reconciles coefficient order, and the full-column-rank affine-pushforward interface used in the affine baseline of [Balcan et al. \(2020\)](#) is absolutely continuous in its affine image. Proposition [A.14](#) therefore rules out every latent dimension while expressly retaining the full-rank and latent-density limitations. Finally, Lemma [A.9](#) and Proposition [A.16](#) recover the bounded-joint- density comparison without adding that premise to the main theorem.

References

BALCAN, M.-F., DICK, T. and PEGDEN, W. (2020). Semi-bandit optimization in the dispersed setting. *Proceedings of Machine Learning Research* **124**. ArXiv:1904.09014. [6](#), [23](#), [24](#)

A Proof Details

A.1 Deterministic Midpoint Localization

Lemma A.1 (Uniform polynomial derivative envelope). *Fix an integer $d \geq 1$, a radius $R > 0$, and a compact set $\Theta \subseteq \mathbb{R}$, and define ϕ_α , B , and $L_{d,R,\Theta}$ as defined above. Then every $\alpha \in [-R, R]^d$ and $\theta \in \Theta$ satisfy*

$$|\partial_\theta \phi_\alpha(\theta)| \leq dB^{d-1} + R \sum_{k=1}^{d-1} kB^{k-1}.$$

Consequently,

$$L_{d,R,\Theta} \leq dB^{d-1} + R \sum_{k=1}^{d-1} kB^{k-1} < \infty.$$

For $d = 1$, the sum is empty and $L_{1,R,\Theta} = 1$.

Proof. Compactness of Θ makes $\sup_{\theta \in \Theta} |\theta|$ finite, so B is finite and at least one. Differentiating gives

$$\partial_\theta \phi_\alpha(\theta) = d\theta^{d-1} + \sum_{k=1}^{d-1} k\alpha_k \theta^{k-1}.$$

Since $|\alpha_k| \leq R$ and $|\theta| \leq B$, the triangle inequality gives, term by term,

$$\begin{aligned} |\partial_\theta \phi_\alpha(\theta)| &\leq d|\theta|^{d-1} + \sum_{k=1}^{d-1} k|\alpha_k||\theta|^{k-1} \\ &\leq dB^{d-1} + R \sum_{k=1}^{d-1} kB^{k-1}. \end{aligned}$$

The right-hand side is a finite sum of finite quantities and is uniform over the coefficient cube and Θ . Taking the defining supremum proves the bound and finiteness of $L_{d,R,\Theta}$. If $d = 1$, then $\phi_\alpha(\theta) = \theta + \alpha_0$ and $\partial_\theta \phi_\alpha \equiv 1$, proving the exact identity. $\square$

Proposition A.1 (Borel root hitting for arbitrary interval endpoints). *Fix an integer $d \geq 1$, a radius $R > 0$, and a compact set $\Theta \subseteq \mathbb{R}$, with ϕ_α and Z_α as in the setup. For every $I \in \mathcal{I}(\Theta)$, under any open, closed, or half-open endpoint convention,*

$$\mathcal{E}_I := \{\alpha \in [-R, R]^d : Z_\alpha \cap I \neq \emptyset\}$$

is a Borel subset of $[-R, R]^d$.

Proof. Let $a = \inf I$ and $b = \sup I$. Both are finite, and $a < b$. For $n \geq 1$, set

$$\varepsilon_n := \frac{b-a}{2(n+1)}, \quad a_n := \begin{cases} a, & a \in I, \\ a + \varepsilon_n, & a \notin I, \end{cases} \quad b_n := \begin{cases} b, & b \in I, \\ b - \varepsilon_n, & b \notin I, \end{cases}$$

and $K_n = [a_n, b_n]$. Because $\varepsilon_n < (b-a)/2$, every K_n is a nonempty compact interval contained in I . Moreover, $K_n \subseteq K_{n+1}$ and

$$\bigcup_{n \geq 1} K_n = I.$$

An endpoint is retained in every K_n exactly when it belongs to I , and every interior point is eventually captured. Thus the exhaustion covers all four endpoint conventions.

For $\alpha \in [-R, R]^d$, define

$$F_n(\alpha) := \min_{\theta \in K_n} |\phi_\alpha(\theta)|.$$

The minimum exists because K_n is nonempty and compact and the polynomial is continuous. For any $\alpha, \gamma \in [-R, R]^d$, put

$$\delta := \sup_{\theta \in K_n} |\phi_\alpha(\theta) - \phi_\gamma(\theta)|.$$

Pointwise on K_n , $|\phi_\gamma(\theta)| - \delta \leq |\phi_\alpha(\theta)| \leq |\phi_\gamma(\theta)| + \delta$. Taking minima and then interchanging α and γ gives

$$\begin{aligned} |F_n(\alpha) - F_n(\gamma)| &\leq \sup_{\theta \in K_n} |\phi_\alpha(\theta) - \phi_\gamma(\theta)| \\ &\leq \sum_{k=0}^{d-1} |\alpha_k - \gamma_k| B^k. \end{aligned}$$

Hence F_n is continuous, and

$$\mathcal{E}_{I,n} := \{\alpha \in [-R, R]^d : F_n(\alpha) = 0\}$$

is closed relative to the coefficient cube. Since the minimum is attained, $F_n(\alpha) = 0$ exactly when ϕ_α has a zero in K_n . The compact exhaustion therefore yields

$$\mathcal{E}_I = \bigcup_{n \geq 1} \mathcal{E}_{I,n},$$

a countable union of closed sets. The argument only asserts existence of a zero on each compact set and never chooses a root as a function of α . $\square$

Proposition A.2 (Measurable midpoint intercept slab). *Fix an integer $d \geq 1$, a radius $R > 0$, a compact set $\Theta \subseteq \mathbb{R}$, and $I \in \mathcal{J}(\Theta)$. Under Lemma A.1, define, with an empty sum when $d = 1$,*

$$c_I(\beta) := m_I^d + \sum_{k=1}^{d-1} \alpha_k m_I^k, \quad \beta = (\alpha_1, \dots, \alpha_{d-1}),$$

and

$$J_I(\beta) := \left[-c_I(\beta) - \frac{L_{d,R,\Theta}|I|}{2}, -c_I(\beta) + \frac{L_{d,R,\Theta}|I|}{2} \right].$$

Then c_I and both endpoints of J_I are Borel measurable, $|J_I(\beta)| = L_{d,R,\Theta}|I|$, and

$$\{(a_0, \beta) : a_0 \in J_I(\beta)\} = \left\{ (a_0, \beta) : |a_0 + c_I(\beta)| \leq \frac{L_{d,R,\Theta}|I|}{2} \right\}$$

is Borel. For every $\alpha = (\alpha_0, \beta) \in [-R, R]^d$,

$$Z_\alpha \cap I \neq \emptyset \implies |\phi_\alpha(m_I)| \leq \frac{L_{d,R,\Theta}|I|}{2} \iff \alpha_0 \in J_I(\beta).$$

The implication holds for every endpoint convention and requires neither simple nor transverse roots.

Proof. Again write $a = \inf I$ and $b = \sup I$. Since $a < b$, the midpoint $m_I = (a + b)/2$ lies strictly between a and b , and hence belongs to I for every endpoint convention. If $\theta_* \in I$, the closed segment joining θ_* and m_I lies in $I \subseteq \Theta$, and

$$|m_I - \theta_*| \leq \frac{b - a}{2} = \frac{|I|}{2}.$$

Suppose $\theta_* \in Z_\alpha \cap I$. If $\theta_* = m_I$, the desired bound has left-hand side zero. Otherwise, the one-dimensional mean-value inequality applies on the segment: polynomiality gives continuity and differentiability, and Lemma A.1 bounds the derivative there. Since $\phi_\alpha(\theta_*) = 0$,

$$\begin{aligned} |\phi_\alpha(m_I)| &= |\phi_\alpha(m_I) - \phi_\alpha(\theta_*)| \\ &\leq L_{d,R,\Theta} |m_I - \theta_*| \\ &\leq \frac{L_{d,R,\Theta} |I|}{2}. \end{aligned}$$

This retains the exact midpoint factor.

The additive-intercept form gives

$$\phi_\alpha(m_I) = \alpha_0 + c_I(\beta),$$

so the evaluation bound is equivalent to $\alpha_0 \in J_I(\beta)$. The endpoint difference of J_I is exactly $L_{d,R,\Theta} |I|$. For fixed I , c_I is a polynomial in β , so it and the interval endpoints are continuous. The map $(a_0, \beta) \mapsto |a_0 + c_I(\beta)|$ is continuous, and the slab is the inverse image of the closed interval $[0, L_{d,R,\Theta} |I|/2]$.

When $d = 1$, β is the empty tuple, $c_I(\beta) = m_I$, $L_{1,R,\Theta} = 1$, and

$$J_I = [-m_I - |I|/2, -m_I + |I|/2].$$

Thus the formulas remain literal on the one-point conditioning space. If an included endpoint is a root, the distance bound may be an equality; an excluded endpoint is absent from the event, while arbitrarily endpoint-near roots obey the same weak inequality. The proof never divides by $|\partial_\theta \phi_\alpha(\theta_*)|$ and never uses root uniqueness, so tangent and multiple roots are covered. Together with Proposition A.1, this supplies a Borel root event contained in a jointly Borel intercept slab of exact width $L_{d,R,\Theta} |I|$. $\square$

A.2 Averaged Conditional-Density Integration

Lemma A.2 (Conditional integration over the midpoint slab). *Under Assumption 1, Proposition A.1, and Proposition A.2, fix $\mu \in \mathcal{D}$ and $I \in \mathcal{I}(\Theta)$. Then the root event and midpoint slab are Borel, the function*

$$(a_0, \beta) \mapsto \mathbf{1}\{a_0 \in J_I(\beta)\} f_\mu(a_0 \mid \beta)$$

is nonnegative and jointly measurable, and

$$\Pr_\mu(Z_\alpha \cap I \neq \emptyset) \leq \int \int_{\mathbb{R}} \mathbf{1}\{a_0 \in J_I(\beta)\} f_\mu(a_0 \mid \beta) da_0 \pi_\mu(d\beta).$$

Proof. Proposition A.1 makes $\mathcal{E}_I = \{Z_\alpha \cap I \neq \emptyset\}$ measurable, while Proposition A.2 makes $\mathcal{S}_I = \{(a_0, \beta) : a_0 \in J_I(\beta)\}$ Borel and gives $\mathcal{E}_I \subseteq \mathcal{S}_I$. Hence

$$\Pr_\mu(\mathcal{E}_I) \leq \Pr_\mu(\mathcal{S}_I).$$

The indicator of $\mathcal{S}_I$ is jointly Borel. Multiplying it by the jointly measurable nonnegative density supplied by Assumption 1 gives the asserted integrand. Regular conditional-density disintegration yields

$$\Pr_\mu(\mathcal{S}_I) = \int_\mu \int_{\mathbb{R}} \mathbf{1}\{a_0 \in J_I(\beta)\} f_\mu(a_0 | \beta) da_0 \pi_\mu(d\beta).$$

The density identity is needed only for π_μ -almost every fiber; the remaining fibers contribute zero to the outer integral. The conditional intercept is supported on $[-R, R]$ and its density is extended by zero, so the formula remains exact even if $J_I(\beta)$ extends outside the cube. Tonelli's theorem for this nonnegative jointly measurable integrand also makes the inner integral a measurable extended-valued function of β . $\square$

Lemma A.3 (Averaged slice-cap control). *Under Assumption 1, Proposition A.2, and Lemma A.2, fix $\mu \in \mathcal{D}$ and $I \in \mathcal{I}(\Theta)$. For π_μ -almost every β ,*

$$\int_{\mathbb{R}} \mathbf{1}\{a_0 \in J_I(\beta)\} f_\mu(a_0 | \beta) da_0 \leq K_\mu(\beta) L_{d,R,\Theta} |I|,$$

and

$$\int \int_{\mathbb{R}} \mathbf{1}\{a_0 \in J_I(\beta)\} f_\mu(a_0 | \beta) da_0 \pi_\mu(d\beta) \leq L_{d,R,\Theta} |I| \int K_\mu(\beta) \pi_\mu(d\beta) < \infty.$$

Proof. For fixed μ , the definition of the class envelope gives

$$0 \leq \int K_\mu(\beta) \pi_\mu(d\beta) \leq \bar{\kappa}_{\mathcal{D}} < \infty.$$

Because K_μ is nonnegative, measurable, and possibly extended-valued, this implies $K_\mu(\beta) < \infty$ almost surely. To see this explicitly, let $N_\mu^\infty = \{\beta : K_\mu(\beta) = \infty\}$. For every integer $n \geq 1$,

$$\int K_\mu d\pi_\mu \geq n \pi_\mu(N_\mu^\infty),$$

so finiteness forces $\pi_\mu(N_\mu^\infty) = 0$.

Intersect this full-measure set with the full-measure set on which $f_\mu(\cdot | \beta)$ is the conditional density and $K_\mu(\beta)$ is its Lebesgue essential supremum. On the resulting full-measure set,

$$f_\mu(a_0 | \beta) \leq K_\mu(\beta) \quad \text{for Lebesgue-almost every } a_0.$$

Proposition A.2 gives $|J_I(\beta)| = L_{d,R,\Theta} |I| < \infty$. Therefore

$$\begin{aligned} \int_{\mathbb{R}} \mathbf{1}\{a_0 \in J_I(\beta)\} f_\mu(a_0 | \beta) da_0 &= \int_{J_I(\beta)} f_\mu(a_0 | \beta) da_0 \\ &\leq K_\mu(\beta) |J_I(\beta)| \\ &= K_\mu(\beta) L_{d,R,\Theta} |I|. \end{aligned}$$

Null fibers, including a fiber assigned $K_\mu = \infty$, contribute nothing to the outer integral. By Lemma A.2 and Tonelli, the left-hand inner integral is measurable and nonnegative. Integrating the last display against the arbitrary Borel probability π_μ proves the second inequality. This uses no density of π_μ and never replaces $K_\mu(\beta)$ by an essential supremum over β . $\square$

Proposition A.3 (Uniform root hitting under an averaged intercept envelope). *Under Assumption 1, Proposition A.1, Proposition A.2, Lemma A.2, and Lemma A.3, every $\mu \in \mathcal{D}$ and $I \in \mathcal{I}(\Theta)$ satisfy*

$$\Pr_{\mu}(Z_{\alpha} \cap I \neq \emptyset) \leq L_{d,R,\Theta}|I| \int K_{\mu}(\beta) \pi_{\mu}(d\beta) \leq L_{d,R,\Theta}|I|\bar{\kappa}_{\mathcal{D}}.$$

Consequently,

$$C_{\mathcal{D}} \leq L_{d,R,\Theta}\bar{\kappa}_{\mathcal{D}} < \infty.$$

Proof. Lemma A.2 bounds the root probability by the nonnegative double integral over the midpoint slab, and Lemma A.3 bounds that integral by the first right-hand side. For the fixed law μ ,

$$\int K_{\mu}(\beta) \pi_{\mu}(d\beta) \leq \sup_{\nu \in \mathcal{D}} \int K_{\nu}(\beta) \pi_{\nu}(d\beta) = \bar{\kappa}_{\mathcal{D}}.$$

Thus each fiber cap is integrated before the class supremum is applied. Since $|I| > 0$, division by $|I|$ gives

$$\sup_{I \in \mathcal{I}(\Theta)} \frac{\Pr_{\mu}(Z_{\alpha} \cap I \neq \emptyset)}{|I|} \leq L_{d,R,\Theta}\bar{\kappa}_{\mathcal{D}}.$$

Taking the outer supremum over $\mu \in \mathcal{D}$ proves the claim. Finiteness follows from Lemma A.1 and Assumption 1.

When $d = 1$, β is the empty tuple and π_{μ} is the point mass on its one-point space. Proposition A.2 gives $L_{1,R,\Theta} = 1$ and a constant interval J_I of length $|I|$; the outer integrals reduce to evaluation at one point. Thus the argument is the ordinary one-dimensional density bound and introduces no hidden positive-dimensional marginal condition. The preceding results also show explicitly that zero extension, unbounded slice caps, singular or discrete higher-coordinate marginals, and every endpoint convention leave the exact coefficient-one conclusion unchanged. $\square$

A.3 Random-Intercept Conditional Kernels

Lemma A.4 (Measurable conditional kernel for an affine uniform intercept). *Let $R > 0$, let β have an arbitrary Borel probability law π on the higher-coordinate cube, with the one-point convention when $d = 1$, and let $U \sim \text{Unif}[-1, 1]$ be independent of β . Let G, ρ be measurable real-valued functions of β and put $Y = G(\beta) + \rho(\beta)U$. Suppose H is measurable, $\pi(H) = 1$, and $\rho(\beta) > 0$ for every $\beta \in H$. Then Y has a jointly measurable regular conditional density $f(a | \beta)$ given β such that, for every $\beta \in H$,*

$$f(a | \beta) = \frac{\mathbf{1}\{|a - G(\beta)| \leq \rho(\beta)\}}{2\rho(\beta)}, \quad \text{ess sup}_{a \in \mathbb{R}} f(a | \beta) = \frac{1}{2\rho(\beta)}.$$

If also $|G(\beta)| + \rho(\beta) \leq R$ on H , the density version can be chosen to vanish outside $[-R, R]$ on every fiber.

Proof. Define the fallback density

$$g_R(a) := \frac{\mathbf{1}\{|a| \leq R\}}{2R}$$

and the piecewise function

$$f(a | \beta) = \begin{cases} \frac{\mathbf{1}\{|a - G(\beta)| \leq \rho(\beta)\}}{2\rho(\beta)}, & \beta \in H, \\ g_R(a), & \beta \notin H. \end{cases}$$

There is no quotient on H^c , so this definition contains no hidden 0/0 convention. The set

$$\{(a, \beta) : |a - G(\beta)| \leq \rho(\beta)\}$$

is measurable, as are H, G , and ρ . Strict positivity on H makes f jointly measurable. Its integral in a equals one on H because the indicated interval has length $2\rho(\beta)$, and equals one on H^c by the definition of g_R . Thus

$$Q(A \mid \beta) := \int_A f(a \mid \beta) da$$

is a measurable probability kernel.

Fix Borel sets $A \subseteq \mathbb{R}$ and B_0 in the higher-coordinate space. Independence gives the product-law identity

$$\Pr(Y \in A, \beta \in B_0) = \int_{B_0} \frac{1}{2} \int_{-1}^1 \mathbf{1}\{G(b) + \rho(b)u \in A\} du \pi(db).$$

For $b \in H$, the positive-width substitution $a = G(b) + \rho(b)u$ gives, for every nonnegative Borel function h ,

$$\frac{1}{2} \int_{-1}^1 h(G(b) + \rho(b)u) du = \int_{\mathbb{R}} h(a) \frac{\mathbf{1}\{|a - G(b)| \leq \rho(b)\}}{2\rho(b)} da.$$

Consequently, the inner integral over A equals $\int_A f(a \mid b) da$ on H . Since H^c has π -measure zero, replacing the inner integral there by the fallback kernel changes nothing, and

$$\Pr(Y \in A, \beta \in B_0) = \int_{B_0} \int_A f(a \mid b) da \pi(db).$$

This rectangle identity proves that Q is a regular conditional law of the actual Y given the actual β . It integrates against π itself and needs no Lebesgue density of π .

For $b \in H$, the displayed density equals the positive constant $1/[2\rho(b)]$ on an interval of positive length $2\rho(b)$ and is zero off that interval. Its essential supremum is therefore exactly $1/[2\rho(b)]$. A globally measurable cap version is

$$K(b) = \begin{cases} 1/[2\rho(b)], & b \in H, \\ 1/(2R), & b \notin H. \end{cases}$$

Finally, if $|G| + \rho \leq R$ on H , then $[G(b) - \rho(b), G(b) + \rho(b)] \subseteq [-R, R]$ there, while g_R is also supported on that interval. Hence the selected version vanishes outside $[-R, R]$ on every fiber. $\square$

Proposition A.4 (Reciprocal-width envelope for random intercepts). *Under Assumption 2 and Lemma A.4, every $\mu \in \mathcal{D}_{\text{RI}}$ is supported on $[-R, R]^d$ and has a jointly measurable regular conditional density, extended by zero outside $[-R, R]$, satisfying for π_μ -almost every β*

$$f_\mu(a_0 \mid \beta) = \frac{\mathbf{1}\{|a_0 - G_\mu(\beta)| \leq \rho_\mu(\beta)\}}{2\rho_\mu(\beta)}, \quad K_\mu(\beta) = \frac{1}{2\rho_\mu(\beta)}.$$

Moreover, K_μ has a measurable version and

$$\int K_\mu(\beta) \pi_\mu(d\beta) = \frac{1}{2} \int \rho_\mu(\beta)^{-1} \pi_\mu(d\beta), \quad \bar{\kappa}_{\mathcal{D}_{\text{RI}}} = \frac{1}{2} M_{\text{RI}} < \infty.$$

Thus $\mathcal{D}_{\text{RI}}$ satisfies every component of Assumption 1.

Proof. Fix $\mu \in \mathcal{D}_{\text{RI}}$ and define

$$H_\mu := \{\beta : \rho_\mu(\beta) > 0, |G_\mu(\beta)| + \rho_\mu(\beta) \leq R\}.$$

This set is measurable, and Assumption 2 gives $\pi_\mu(H_\mu) = 1$. Apply Lemma A.4 with $H = H_\mu$. It produces a jointly measurable conditional density with the asserted formula and cap. Its support clause gives a version vanishing outside $[-R, R]$ on every fiber, and its piecewise cap gives a measurable essential-supremum version on every fiber.

The higher-coordinate vector already lies in $[-R, R]^{d-1}$. On the probability-one event $\{\beta \in H_\mu, |U_\mu| \leq 1\}$,

$$|\alpha_0| \leq |G_\mu(\beta)| + \rho_\mu(\beta)|U_\mu| \leq |G_\mu(\beta)| + \rho_\mu(\beta) \leq R.$$

Therefore the actual coefficient law is supported on $[-R, R]^d$.

The complement of H_μ is null, so neither the fallback cap nor any arbitrary value assigned to ρ_μ^{-1} there contributes. Hence

$$\begin{aligned} \int K_\mu(\beta) \pi_\mu(d\beta) &= \frac{1}{2} \int_{H_\mu} \rho_\mu(\beta)^{-1} \pi_\mu(d\beta) \\ &= \frac{1}{2} \int \rho_\mu(\beta)^{-1} \pi_\mu(d\beta). \end{aligned}$$

Only after this per-law integration do we take the class supremum:

$$\begin{aligned} \bar{\kappa}_{\mathcal{D}_{\text{RI}}} &= \sup_{\mu \in \mathcal{D}_{\text{RI}}} \int K_\mu d\pi_\mu \\ &= \frac{1}{2} \sup_{\mu \in \mathcal{D}_{\text{RI}}} \int \rho_\mu^{-1} d\pi_\mu = \frac{1}{2} M_{\text{RI}}. \end{aligned}$$

No essential supremum over β has been inserted. When $d = 1$, the higher-coordinate space is the one-point empty-tuple space and all these integrals are evaluations at that point. The kernel, support estimate, and reciprocal-width identity remain literal. Joint measurability of f_μ , measurability of K_μ , and finiteness of the last display verify all components of Assumption 1. $\square$

Proposition A.5 (Uniform root hitting for random intercepts). *Under Assumption 2, Proposition A.4, and Proposition A.3, every $\mu \in \mathcal{D}_{\text{RI}}$ and $I \in \mathcal{I}(\Theta)$ satisfy*

$$\begin{aligned} \Pr_\mu(Z_\alpha \cap I \neq \emptyset) &\leq \frac{L_{d,R,\Theta}|I|}{2} \int \rho_\mu(\beta)^{-1} \pi_\mu(d\beta) \\ &\leq L_{d,R,\Theta}|I| \bar{\kappa}_{\mathcal{D}_{\text{RI}}} \\ &\leq \frac{L_{d,R,\Theta} M_{\text{RI}}}{2} |I|, \end{aligned}$$

and

$$C_{\mathcal{D}_{\text{RI}}} \leq L_{d,R,\Theta} \bar{\kappa}_{\mathcal{D}_{\text{RI}}} \leq \frac{1}{2} L_{d,R,\Theta} M_{\text{RI}} < \infty.$$

Proof. Proposition A.4 proves from the primitive random-intercept assumption that the class is cube-supported and satisfies the full conditional-density and averaged-envelope hypothesis of Proposition A.3. Applying that proposition to $\mathcal{D}_{\text{RI}}$ gives

$$\Pr_\mu(Z_\alpha \cap I \neq \emptyset) \leq L_{d,R,\Theta}|I| \int K_\mu d\pi_\mu.$$

Substitution of the exact per-law identity from Proposition A.4, followed by the definition of M_{RI} , gives

$$\begin{aligned} L_{d,R,\Theta}|I| \int K_\mu d\pi_\mu &= \frac{L_{d,R,\Theta}|I|}{2} \int \rho_\mu^{-1} d\pi_\mu \\ &\leq L_{d,R,\Theta}|I|\bar{\kappa}_{\mathcal{D}_{\text{RI}}} \\ &\leq \frac{L_{d,R,\Theta}M_{\text{RI}}}{2}|I|. \end{aligned}$$

The class conclusion follows in the same way. The fixed law's reciprocal width is integrated before the class supremum is used. Since Proposition A.3 is already uniform over all positive-length intervals, no interval-dependent exceptional set or probability conversion arises. The one-point $d = 1$ marginal is covered by the same substitution. $\square$

Proposition A.6 (Exact fixed-width certificate). *Under Assumption 2, Proposition A.4, and Proposition A.5, suppose $\mu \in \mathcal{D}_{\text{RI}}$ has $\rho_\mu(\beta) = r_\mu$ for π_μ -almost every β , where $r_\mu > 0$. Then*

$$K_\mu(\beta) = \frac{1}{2r_\mu} \quad \text{for } \pi_\mu\text{-almost every } \beta, \quad \int K_\mu d\pi_\mu = \frac{1}{2r_\mu},$$

and, for every $I \in \mathcal{I}(\Theta)$,

$$\Pr_\mu(Z_\alpha \cap I \neq \emptyset) \leq \frac{L_{d,R,\Theta}}{2r_\mu}|I|.$$

If $\mathcal{F}$ is a nonempty class of such fixed-width members satisfying the same support condition and $\sup_{\mu \in \mathcal{F}} r_\mu^{-1} < \infty$, then

$$\bar{\kappa}_{\mathcal{F}} = \frac{1}{2} \sup_{\mu \in \mathcal{F}} r_\mu^{-1}, \quad C_{\mathcal{F}} \leq \frac{L_{d,R,\Theta}}{2} \sup_{\mu \in \mathcal{F}} r_\mu^{-1} < \infty.$$

Proof. On the full-measure set where the fixed-width identity and Assumption 2 hold, Proposition A.4 gives

$$K_\mu(\beta) = \frac{1}{2\rho_\mu(\beta)} = \frac{1}{2r_\mu}.$$

Integration with respect to the probability law π_μ proves the cap integral, and substitution into Proposition A.5 proves the per-law root bound. For a class $\mathcal{F}$, take the supremum only after the per-law calculation:

$$\bar{\kappa}_{\mathcal{F}} = \sup_{\mu \in \mathcal{F}} \frac{1}{2r_\mu} = \frac{1}{2} \sup_{\mu \in \mathcal{F}} r_\mu^{-1}.$$

The stated reciprocal-width condition is exactly the class-uniform integrability condition specialized to fixed widths. Applying Proposition A.3 to $\mathcal{F}$ gives the class conclusion. For $d = 1$, r_μ is the width of the unique empty-tuple fiber and all equations remain unchanged. Together with Propositions A.4 and A.5, this proves the conditional density, exact cap, averaged budget, root bound, and fixed-width clauses without any higher-marginal density assumption. $\square$

A.4 Exact Heteroscedastic Sheet and Conditional Law

Proposition A.7 (Exact support of the heteroscedastic sheet). *Fix $d \geq 3$, $R > 0$, and $q \in (0, 1)$, and let μ_q be the explicit witness law from the setup. Define*

$$T_q = [-R/2, R/2] \times [-1, 1]$$

and

$$F_q : T_q \longrightarrow \mathbb{R}^d, \quad F_q(z, u) = (\rho_q(z)u, z, z^2/R, 0, \dots, 0).$$

When $d = 3$, the trailing list of zeros is empty. Then

$$\text{supp}(\mu_q) = F_q(T_q) = S_q,$$

where

$$S_q = \{(a_0, z, z^2/R, 0, \dots, 0) : |z| \leq R/2, |a_0| \leq \rho_q(z)\}.$$

The identity includes the collapsed fiber at $z = 0$ and the fibers at $z = \pm R/2$.

Proof. Since $q > 0$, the map $z \mapsto |z|^q$, and hence ρ_q , is continuous at every z , including zero. Since $R > 0$, all other coordinates of F_q are continuous. Thus F_q is continuous on the compact rectangle T_q and $F_q(T_q)$ is compact, hence closed.

For $(z, u) \in T_q$, $|\rho_q(z)u| \leq \rho_q(z)$, so $F_q(T_q) \subseteq S_q$. Conversely, take a point of S_q . If $z \neq 0$, then $\rho_q(z) > 0$ and

$$u = \frac{a_0}{\rho_q(z)} \in [-1, 1]$$

maps to that point. If $z = 0$, then $\rho_q(0) = 0$ and the sheet inequality forces $a_0 = 0$; choosing $u = 0$ again gives the point. Thus

$$F_q(T_q) = S_q.$$

Independence and the uniform laws of Z and U make their joint law normalized two-dimensional Lebesgue measure on T_q . Every nonempty relatively open subset of T_q , including one meeting an edge or corner, has positive probability, so the support of (Z, U) is exactly T_q . Moreover, $\alpha^{(q)} = F_q(Z, U)$, and therefore $\mu_q = (F_q)_\# \mathcal{L}(Z, U)$.

The pushforward is concentrated on the closed set $F_q(T_q)$, giving one support inclusion. For the reverse inclusion, fix $y = F_q(z, u)$ and an open neighborhood $O \ni y$. Continuity makes $F_q^{-1}(O)$ a relatively open subset of T_q containing (z, u) , so it has positive joint probability and

$$\mu_q(O) = \Pr(F_q(Z, U) \in O) > 0.$$

Thus every image point belongs to the topological support.

At $z = 0$, every $(0, u)$ maps to $(0, 0, 0, \dots, 0)$. Although $\Pr(Z = 0) = 0$, every neighborhood of this point has positive pushforward mass by the preimage argument, so the collapsed point remains in support. At $z = \pm R/2$, relative neighborhoods of the latent boundary also have positive mass; hence the full endpoint fibers, including their $u = \pm 1$ endpoints, remain in support. $\square$

Proposition A.8 (Cube support and almost-sure positive width). *Fix $d \geq 3$, $R > 0$, and $q \in (0, 1)$. Under Proposition A.7,*

$$S_q \subseteq [-R, R]^d, \quad \mu_q([-R, R]^d) = 1, \quad \Pr(\rho_q(Z) > 0) = 1.$$

These conclusions include $d = 3$, the collapsed support point at $z = 0$, and both endpoint fibers $z = \pm R/2$.

Proof. For $|z| \leq R/2$, put $t = 2|z|/R$. Since $R > 0$, $t \in [0, 1]$, and $q > 0$ gives $t^q \in [0, 1]$. Hence

$$0 \leq \rho_q(z) = \frac{R}{2} t^q \leq \frac{R}{2}.$$

Every point of S_q satisfies

$$|a_0| \leq \rho_q(z) \leq R/2 \leq R, \quad |\alpha_1| = |z| \leq R/2 \leq R,$$

and

$$|\alpha_2| = \frac{z^2}{R} \leq \frac{(R/2)^2}{R} = \frac{R}{4} \leq R.$$

All remaining coordinates vanish. This proves $S_q \subseteq [-R, R]^d$, and Proposition A.7 gives cube support of the actual law.

Because $R > 0$ and $q > 0$,

$$\rho_q(z) = 0 \iff z = 0.$$

The uniform law on the nondegenerate interval $[-R/2, R/2]$ has no atoms, so $\Pr(Z = 0) = 0$ and the almost-sure assertion follows. This does not remove the origin from topological support.

When $d = 3$, the only coordinates are $a_0, z, z^2/R$, and the same bounds apply with no trailing coordinates. At $z = \pm R/2$, $\rho_q(z) = R/2$ and $z^2/R = R/4$; at $z = 0$, all displayed coordinates vanish. The upper restriction $q < 1$ is not needed for these support claims; it is used in the cap integral below. $\square$

Lemma A.5 (Recovery of the latent coordinate and inherited independence). *Fix $d \geq 3$, $R > 0$, and $q \in (0, 1)$. For a higher-coordinate vector $b \in [-R, R]^{d-1}$, let b_1 denote its first coordinate and define*

$$G_q(b) = 0, \quad \tilde{\rho}_q(b) = \frac{R}{2} \left(\frac{2|b_1|}{R} \right)^q, \quad H_q = \{b : 0 < |b_1| \leq R/2\}.$$

Then $G_q, \tilde{\rho}_q$, and H_q are measurable,

$$Z = \text{pr}_1(\beta_q(Z)), \quad \sigma(\beta_q(Z)) = \sigma(Z),$$

U is independent of $\beta_q(Z)$, $\pi_{\mu_q}(H_q) = 1$, $\tilde{\rho}_q > 0$ on H_q , and

$$\rho_q(Z)U = G_q(\beta_q(Z)) + \tilde{\rho}_q(\beta_q(Z))U.$$

Thus all conditions of Lemma A.4 hold for this construction.

Proof. The condition $d \geq 3$ ensures that the higher-coordinate vector has a first coordinate. By definition,

$$\beta_q(Z) = (Z, Z^2/R, 0, \dots, 0),$$

so projection onto that coordinate recovers Z exactly. Since $\beta_q(Z)$ is a measurable function of Z , $\sigma(\beta_q(Z)) \subseteq \sigma(Z)$; recovery gives the reverse inclusion. The signed linear coordinate prevents any ambiguity from the quadratic coordinate.

For Borel sets $A \subseteq [-1, 1]$ and $C \subseteq [-R, R]^{d-1}$, independence of U and Z gives

$$\begin{aligned} \Pr(U \in A, \beta_q(Z) \in C) &= \Pr(U \in A, Z \in \beta_q^{-1}(C)) \\ &= \Pr(U \in A) \Pr(Z \in \beta_q^{-1}(C)) \\ &= \Pr(U \in A) \Pr(\beta_q(Z) \in C). \end{aligned}$$

Thus U is independent of the actual conditioning vector.

The functions above are continuous and H_q is Borel. Along the actual higher-coordinate curve,

$$\tilde{\rho}_q(\beta_q(z)) = \frac{R}{2} \left(\frac{2|z|}{R} \right)^q = \rho_q(z).$$

This proves the representation. Also,

$$\pi_{\mu_q}(H_q) = \Pr(0 < |Z| \leq R/2) = 1,$$

because Z has no atom at zero. For $b \in H_q$, positivity of R, q and $b_1 \neq 0$ imply $\tilde{\rho}_q(b) > 0$, and $|G_q(b)| + \tilde{\rho}_q(b) \leq R/2 \leq R$. Thus measurability, representation, independence, full measure, positive width, and support are all established before Lemma A.4 is used. No reciprocal-width integrability is asserted here. $\square$

Proposition A.9 (Conditional density and exact heteroscedastic cap). *Fix $d \geq 3$, $R > 0$, and $q \in (0, 1)$. Under Lemma A.5 and Lemma A.4, the actual intercept $\alpha_0^{(q)} = \rho_q(Z)U$ has a jointly measurable regular conditional density given $\beta_q(Z)$. One valid version satisfies, for every $0 < |z| \leq R/2$,*

$$f_{\mu_q}(a_0 \mid \beta_q(z)) = \frac{\mathbf{1}\{|a_0| \leq \rho_q(z)\}}{2\rho_q(z)},$$

and its exact Lebesgue essential-supremum cap is

$$K_{\mu_q}(\beta_q(z)) = \frac{1}{2\rho_q(z)} = \frac{1}{R(2|z|/R)^q}.$$

At $z = 0$, neither formula is asserted; the selected conditional version uses a genuine fallback density on that π_{μ_q} -null fiber.

Proof. Lemma A.5 proves that Lemma A.4 applies with

$$\beta = \beta_q(Z), \quad G = G_q, \quad \rho = \tilde{\rho}_q, \quad H = H_q, \quad Y = \alpha_0^{(q)}.$$

The noise is therefore independent of the actual conditioning vector, not merely of an unrecovered latent variable. A valid density version is

$$f_{\mu_q}(a_0 \mid b) = \begin{cases} \frac{\mathbf{1}\{|a_0| \leq \tilde{\rho}_q(b)\}}{2\tilde{\rho}_q(b)}, & b \in H_q, \\ \frac{\mathbf{1}\{|a_0| \leq R\}}{2R}, & b \notin H_q. \end{cases}$$

The second line is a probability density because $R > 0$. It defines a measurable version on the null complement without any quotient involving $\tilde{\rho}_q = 0$.

For $0 < |z| \leq R/2$, $\beta_q(z) \in H_q$ and Lemma A.5 turns the first line into the asserted density. It is the positive constant $1/[2\rho_q(z)]$ on an interval of positive length $2\rho_q(z)$ and zero elsewhere, so its essential supremum is exactly that constant. Since

$$2\rho_q(z) = R \left(\frac{2|z|}{R} \right)^q,$$

the cap has the displayed form.

At $z = \pm R/2$, the denominator is R and the cap is exactly $1/R$. At $z = 0$, Proposition A.7 retains the collapsed point in topological support, but $\pi_{\mu_q}(\{\beta_q(0)\}) = \Pr(Z = 0) = 0$. A regular conditional law is not determined there, and the fallback is a legal version choice rather than a positive-mass conditional-uniform assertion. Combining this conditional law with Propositions A.7 and A.8 gives the exact support, cube, positive-width, latent-recovery, and nonzero-fiber cap conclusions for the witness, without evaluating a reciprocal-width integral at the null fiber. $\square$

A.5 Unbounded Slice Caps with a Finite Average

Lemma A.6 (Positive-mass blowup of the heteroscedastic slice cap). *Fix $d \geq 3$, $R > 0$, and $q \in (0, 1)$. Under Proposition A.9, for every $M > 0$,*

$$\pi_{\mu_q}\{\beta : K_{\mu_q}(\beta) > M\} > 0.$$

Consequently,

$$\operatorname{ess\,sup}_{\beta \sim \pi_{\mu_q}} K_{\mu_q}(\beta) = \infty.$$

Both conclusions are unchanged by any legal conditional-density fallback on the marginal-null fiber $\beta_q(0)$.

Proof. For $z \in [-R/2, R/2]$, define the dimensionless coordinate

$$t(z) := \frac{2|z|}{R} \in [0, 1].$$

Fix $M > 0$ and put

$$\delta_M := \min \left\{ \frac{1}{2}, (2RM)^{-1/q} \right\}.$$

Because R, q, M are positive, $0 < \delta_M \leq 1/2$, and

$$\delta_M^q \leq \frac{1}{2RM}, \quad \frac{1}{R\delta_M^q} \geq 2M.$$

Let b_1 be the first coordinate of a higher-coordinate vector b and define the Borel set

$$A_M := \left\{ b : 0 < \frac{2|b_1|}{R} < \delta_M \right\}.$$

Lemma A.5 gives $(\beta_q(z))_1 = z$. Since Z is uniform on an interval of length R ,

$$\begin{aligned} \pi_{\mu_q}(A_M) &= \Pr \left(0 < \frac{2|Z|}{R} < \delta_M \right) \\ &= \frac{1}{R} \left| \left(-\frac{R\delta_M}{2}, \frac{R\delta_M}{2} \right) \setminus \{0\} \right| = \delta_M > 0. \end{aligned}$$

For every actual fiber $b = \beta_q(z) \in A_M$, $z \neq 0$, and Proposition A.9 gives

$$K_{\mu_q}(b) = \frac{1}{Rt(z)^q} > \frac{1}{R\delta_M^q} \geq 2M > M.$$

Thus the cap exceeds M on a set of positive marginal measure.

A nonnegative measurable function has finite essential supremum only if some finite M bounds it outside a null set, which the preceding result rules out. The set A_M expressly excludes $b_1 = 0$, while $\pi_{\mu_q}\{\beta_q(0)\} = 0$, so a fallback modification cannot affect the conclusion. At $z = \pm R/2$, the cap is the finite value $1/R$; the blowup comes solely from positive-mass punctured neighborhoods of zero. The strict condition $q > 0$ is exactly what makes the cap diverge there. $\square$

Lemma A.7 (Exact averaged cap of the power-width witness). *Fix $d \geq 3$, $R > 0$, and $q \in (0, 1)$. Under Proposition A.9,*

$$\int K_{\mu_q}(\beta) \pi_{\mu_q}(d\beta) = \frac{1}{R(1-q)} < \infty.$$

For every dimensionless cutoff $\delta \in (0, 1]$, the cap mass on $t(z) \geq \delta$ is

$$\frac{1 - \delta^{1-q}}{R(1-q)},$$

and the omitted central mass is exactly $\delta^{1-q}/[R(1-q)]$, which vanishes as $\delta \downarrow 0$.

Proof. Retain $t(z) = 2|z|/R$ and define

$$\widehat{K}_q(z) := \begin{cases} \frac{1}{Rt(z)^q}, & z \neq 0, \\ 0, & z = 0. \end{cases}$$

Proposition A.9 gives $\widehat{K}_q(z) = K_{\mu_q}(\beta_q(z))$ for every $z \neq 0$. They may differ at zero, but the uniform law assigns zero probability there. Because $\pi_{\mu_q} = (\beta_q)_\# \mathcal{L}(Z)$,

$$\int K_{\mu_q} d\pi_{\mu_q} = \mathbb{E}[\widehat{K}_q(Z)].$$

This remains true even if a selected null-fiber cap is infinite: a nonnegative extended-valued function supported on a null set has integral zero.

For integers $n \geq 1$, define

$$g_n(z) := \widehat{K}_q(z) \mathbf{1}\{t(z) \geq 1/n\}.$$

Each g_n is nonnegative and Borel. Its cutoff set increases with n , so $g_n(z) \uparrow \widehat{K}_q(z)$ for every z , including zero. By monotone convergence and the exact uniform density $1/R$,

$$\mathbb{E}[\widehat{K}_q(Z)] = \lim_{n \rightarrow \infty} \frac{1}{R} \int_{-R/2}^{R/2} g_n(z) dz.$$

For each n , symmetry and the substitution $t = 2z/R$, $dz = (R/2)dt$, yield

$$\begin{aligned} \frac{1}{R} \int_{-R/2}^{R/2} g_n(z) dz &= \frac{2}{R} \int_{R/(2n)}^{R/2} \frac{1}{R(2z/R)^q} dz \\ &= \frac{1}{R} \int_{1/n}^1 t^{-q} dt \\ &= \frac{1 - n^{-(1-q)}}{R(1-q)}. \end{aligned}$$

Every normalization is visible: the first $1/R$ is the density of Z , the second is part of the cap, symmetry contributes two, and $dz = (R/2)dt$ leaves the single prefactor $1/R$.

Since $q < 1$, $1 - q > 0$ and $n^{-(1-q)} \rightarrow 0$, proving the exact integral. Replacing $1/n$ by any $\delta \in (0, 1]$ gives the cutoff mass; subtracting it from the full integral gives the omitted mass.

The point $z = 0$ is excluded only from the pointwise formula and remains a null point in the integration domain. At $z = \pm R/2$, $t = 1$ and the cap is $1/R$, so no endpoint singularity occurs. At the excluded lower boundary $q = 0$, the mean remains finite but Lemma A.6 fails; at the excluded upper boundary $q = 1$, the logarithmic integral $\int_0^1 t^{-1} dt$ diverges. The degree condition $d \geq 3$ identifies the coefficient vector but introduces no normalization factor. $\square$

Proposition A.10 (Root hitting for the heteroscedastic witness). *Fix $d \geq 3$, $R > 0$, $q \in (0, 1)$, and compact $\Theta \subseteq \mathbb{R}$. Under Proposition A.8, Proposition A.9, Lemma A.7, and Proposition A.3, the singleton class $\{\mu_q\}$ satisfies Assumption 1, with*

$$\bar{\kappa}_{\{\mu_q\}} = \int K_{\mu_q}(\beta) \pi_{\mu_q}(d\beta) = \frac{1}{R(1-q)}.$$

Consequently, for every $I \in \mathcal{I}(\Theta)$,

$$\Pr_{\mu_q}(Z_\alpha \cap I \neq \emptyset) \leq \frac{L_{d,R,\Theta}|I|}{R(1-q)}, \quad C_{\{\mu_q\}} \leq \frac{L_{d,R,\Theta}}{R(1-q)}.$$

Proof. Proposition A.8 proves cube support. Proposition A.9 supplies a jointly measurable regular conditional-density version of the actual intercept given the actual higher-coordinate vector, together with a measurable essential-supremum cap. On the full-measure set of nonzero fibers, that cap has the formula used in Lemma A.7, which proves the cap integral is finite and has the displayed exact value. A kernel or cap modification at $\beta_q(0)$ changes neither the integral nor any conditional identity on the full-measure set.

Thus the singleton class satisfies every component of Assumption 1; none is assumed for the witness. Conditional-density and cap measurability come from Proposition A.9, and finite averaged envelope comes from Lemma A.7. A supremum over a singleton is its sole value.

The conditions of Proposition A.3 are now verified. Applying its per-law conclusion and substituting the exact cap integral gives the interval inequality. Division by $|I| > 0$ and the defining interval supremum give the bound on $C_{\{\mu_q\}}$. There is no class-supremum loss and no pointwise-in- β cap assumption. Finiteness follows from $R > 0$, $1 - q > 0$, and Lemma A.1. Combining this proposition with Lemma A.6 proves simultaneously that the slice cap is essentially unbounded, its mean is exactly finite, and the root-hitting constant has the stated exact dependence, with no cutoff remainder or hidden factor. $\square$

A.6 Geometry Beyond the Full-Rank Affine-Latent Baseline

Proposition A.11 (Exact affine hull of the witness sheet). *Fix $d \geq 3$, $R > 0$, and $q \in (0, 1)$. Under Proposition A.7, define*

$$E_d := \{(a, z, w, 0, \dots, 0) : a, z, w \in \mathbb{R}\} \subseteq \mathbb{R}^d.$$

Then

$$\text{aff}(S_q) = E_d, \quad \dim \text{aff}(S_q) = 3.$$

For $d = 3$, $E_d = \mathbb{R}^3$.

Proof. Every point of S_q has only its first three coordinates possibly nonzero, and the collapsed point

$$o = (0, 0, 0, 0, \dots, 0)$$

belongs to S_q . Hence $\text{aff}(S_q)$ is a linear subspace contained in E_d .

At either endpoint $z = \pm R/2$,

$$\rho_q(\pm R/2) = \frac{R}{2} \left(\frac{2(R/2)}{R} \right)^q = \frac{R}{2}.$$

Therefore the exact support contains

$$o, \quad p_+ = \left(0, \frac{R}{2}, \frac{R}{4}, 0, \dots, 0\right), \quad p_- = \left(0, -\frac{R}{2}, \frac{R}{4}, 0, \dots, 0\right),$$

and

$$r = \left(\frac{R}{2}, \frac{R}{2}, \frac{R}{4}, 0, \dots, 0 \right).$$

The projections of o, p_+, p_- onto the (α_1, α_2) coordinates are

$$(0, 0), \quad (R/2, R/4), \quad (-R/2, R/4),$$

and the determinant of the two nonzero vectors is

$$\det \begin{pmatrix} R/2 & -R/2 \\ R/4 & R/4 \end{pmatrix} = \frac{R^2}{4} > 0.$$

Thus the curved projection supplies two independent affine directions. More explicitly, the direction space contains

$$p_+ - p_- = Re_1, \quad p_+ + p_- = \frac{R}{2}e_2, \quad r - p_+ = \frac{R}{2}e_0.$$

Because $R > 0$, these span E_d . This proves the reverse containment and the affine-hull identity. For $d = 3$, the same directions span all of $\mathbb{R}^3$. The collapsed origin causes no loss because it belongs to the topological support, while the nonzero directions come from the endpoint fibers. $\square$

Proposition A.12 (Zero three-volume inside the affine hull). *Fix $d \geq 3$, $R > 0$, and $q \in (0, 1)$. Under Proposition A.7, define*

$$D_q := \{(a, z) \in \mathbb{R}^2 : |z| \leq R/2, |a| \leq \rho_q(z)\}$$

and

$$T : D_q \longrightarrow E_d, \quad T(a, z) = (a, z, z^2/R, 0, \dots, 0).$$

Then $T(D_q) = S_q$, the map T is $\sqrt{2}$ -Lipschitz in Euclidean norm, and, under Proposition A.11,

$$\mathcal{H}_{E_d}^3(S_q) = 0.$$

Proof. The definitions give both inclusions directly: each $(a, z) \in D_q$ maps to the displayed sheet, and each sheet point is $T(a, z)$ for its first two coordinates. Hence

$$T(D_q) = S_q.$$

No division by $\rho_q(z)$ occurs, including at $z = 0$.

Take $(a, z), (a', z') \in D_q$. Since $|z|, |z'| \leq R/2$,

$$\left| \frac{z^2 - (z')^2}{R} \right| = \frac{|z - z'| |z + z'|}{R} \leq |z - z'|.$$

Consequently,

$$\begin{aligned} \|T(a, z) - T(a', z')\|_2^2 &= |a - a'|^2 + |z - z'|^2 + \left| \frac{z^2 - (z')^2}{R} \right|^2 \\ &\leq |a - a'|^2 + 2|z - z'|^2 \\ &\leq 2(|a - a'|^2 + |z - z'|^2). \end{aligned}$$

This is the claimed $\sqrt{2}$ -Lipschitz estimate.

Proposition A.8 gives $\rho_q(z) \leq R/2$, so D_q lies in the square $[-R/2, R/2]^2$. A mesh of side length δ covers this square with at most $(\lceil R/\delta \rceil + 1)^2$ squares, each of diameter at most $\sqrt{2}\delta$. The sum of cubed diameters is bounded by

$$(\lceil R/\delta \rceil + 1)^2 (\sqrt{2}\delta)^3,$$

which tends to zero as $\delta \downarrow 0$. Thus $\mathcal{H}^3(D_q) = 0$. An L -Lipschitz map satisfies $\mathcal{H}^s(F(A)) \leq L^s \mathcal{H}^s(A)$: map a fine cover of A , noting that each diameter grows by at most L , and then take the defining infimum and limit. Applying this cover estimate gives

$$\mathcal{H}_{E_d}^3(S_q) = \mathcal{H}_{E_d}^3(T(D_q)) \leq (\sqrt{2})^3 \mathcal{H}^3(D_q) = 0.$$

The regular map here is $T(a, z)$. The latent parametrization $(z, u) \mapsto (\rho_q(z)u, z, z^2/R, 0, \dots, 0)$ is not used; its first coordinate is only Hölder at zero when $q < 1$. This distinction handles the collapsed fiber and the minimal case $d = 3$ without any unsupported regularity assertion. $\square$

Lemma A.8 (Fixed-monic descending-coordinate embedding). *Under Propositions A.11 and A.12, define*

$$\iota_d : \mathbb{R}^d \longrightarrow \mathbb{R}^{d+1}, \quad \iota_d(a_0, a_1, \dots, a_{d-1}) = (1, a_{d-1}, a_{d-2}, \dots, a_0).$$

Then ι_d places the branch coefficient vector in descending full-coefficient order with fixed leading coefficient $\alpha_d = 1$. It is an affine isometry, and

$$\begin{aligned} \text{supp}((\iota_d)_\# \mu_q) &= \iota_d(S_q), \\ \dim \text{aff}(\iota_d(S_q)) &= 3, \quad \mathcal{H}_{\text{aff}(\iota_d(S_q))}^3(\iota_d(S_q)) = 0. \end{aligned}$$

Proof. For $x, y \in \mathbb{R}^d$, the leading coordinates of $\iota_d(x)$ and $\iota_d(y)$ cancel and the remaining coordinates are merely reversed. Thus

$$\|\iota_d(x) - \iota_d(y)\|_2 = \|x - y\|_2.$$

The map is an affine isometry onto the closed fixed-monic hyperplane. Its formula gives the exact ordering bridge from $(\alpha_0, \dots, \alpha_{d-1})$ to $(\alpha_d, \dots, \alpha_0)$ with $\alpha_d = 1$.

Write $\iota_d(x) = c_* + Q_d x$. The isometry gives $Q_d^\top Q_d = I_d$. If a lower-coordinate affine map has linear part B_0 , then its fixed-monic composition has linear part $Q_d B_0$, and

$$(Q_d B_0)^\top (Q_d B_0) = B_0^\top B_0.$$

Thus the translation preserves full column rank and the exact Gram-determinant normalization.

Because ι_d is a homeomorphism onto its closed image, it transports support exactly: every neighborhood of $\iota_d(x)$ with $x \in S_q$ pulls back to a neighborhood of a support point and has positive pushforward mass, while the pushforward is concentrated on the closed set $\iota_d(S_q)$. Affine isometries commute with affine hulls and preserve their dimensions and induced Hausdorff volumes. The remaining claims follow from Propositions A.11 and A.12. $\square$

The affine generalization in Appendix Theorem 18 of Balcan et al. (2020) uses the descending full coefficient vector, a full-column-rank linear part A , affine-image volume proportional to $\sqrt{\det(A^\top A)}$, and an induced density divided by the same factor. In particular, a latent cube $[-R_{\text{lat}}, R_{\text{lat}}]^k$ has image volume $\sqrt{\det(A^\top A)}(2R_{\text{lat}})^k$, and a latent density bounded by κ induces a within-image density bounded by $\kappa/\sqrt{\det(A^\top A)}$. The proof in that source unambiguously uses the full descending vector $(\alpha_d, \dots, \alpha_0)$; this is the convention reconciled by Lemma A.8. The source's no-forced-root premise belongs to its subsequent root-hitting conclusion and is not needed for the representability question considered here. The following proposition derives the exact measure statement directly.

Proposition A.13 (Full-rank affine pushforward geometry). *Let $k \geq 1$. Let $X \in \mathbb{R}^k$ have a probability density p with respect to $\mathcal{L}^k$, with $p \leq \kappa < \infty$ almost everywhere and bounded support. Let*

$$f(x) = b + Ax, \quad A \in \mathbb{R}^{(d+1) \times k}, \quad \text{rank}(A) = k,$$

and put

$$L = b + \text{im}(A), \quad J_A := \sqrt{\det(A^\top A)} > 0.$$

Then $\nu := \mathfrak{f}_\# \mathcal{L}(X)$ is absolutely continuous with respect to $\mathcal{H}_L^k$, with exact density

$$g(b + Ax) = \frac{p(x)}{J_A} \quad \text{for Lebesgue-almost every } x, \quad g \leq \frac{\kappa}{J_A}.$$

Moreover,

$$\text{aff}(\text{supp } \nu) = L, \quad \dim \text{aff}(\text{supp } \nu) = k.$$

Proof. Full column rank makes $\mathfrak{f}$ one-to-one from $\mathbb{R}^k$ onto L , with inverse on L

$$x = A^\dagger(y - b), \quad A^\dagger = (A^\top A)^{-1} A^\top.$$

The k -dimensional volume scaling of A is the Gram determinant: for every Borel set $B_0 \subseteq \mathbb{R}^k$,

$$\mathcal{H}_L^k(b + AB_0) = J_A \mathcal{H}_L^k(B_0).$$

Applying this identity first to simple functions and then to the nonnegative density p gives, for every Borel $C \subseteq L$,

$$\begin{aligned} \nu(C) &= \int_{\mathfrak{f}^{-1}(C)} p(x) dx \\ &= \int_C \frac{p(A^\dagger(y - b))}{J_A} d\mathcal{H}_L^k(y). \end{aligned}$$

This proves the exact density, its bound, and within-image absolute continuity.

It remains to justify the support dimension because bounded density does not require p to be positive throughout a bounding cube. Every Borel probability on Euclidean space gives full mass to its topological support: the complement is the union of a countable base of open sets of measure zero. If $H = \text{aff}(\text{supp } \nu)$ were a proper affine subspace of L , then $\mathcal{H}_L^k(H) = 0$ but $\nu(H) = 1$, contradicting the absolute continuity just proved. Hence $H = L$, with no full-support or positive lower-density assumption on X . $\square$

Proposition A.14 (Exclusion of every affine-latent dimension). *Fix $d \geq 3$, $R > 0$, and $q \in (0, 1)$. Under Proposition A.7, Lemma A.8, and Proposition A.13, the law $(\iota_d)_\# \mu_q$ is not the affine pushforward of any bounded-support, bounded-Lebesgue-density latent probability under a full-column-rank affine map into $\mathbb{R}^{d+1}$. Consequently, μ_q is not an affine image of such a latent law in the fixed-monic affine convention of Balcan et al. (2020).*

Proof. If μ_q had such a representation in ascending lower-coordinate order, composition with ι_d would give one in descending full-coordinate order. Lemma A.8 shows that this composition preserves full column rank and the Gram determinant. It therefore suffices to exclude the descending-order representations.

Suppose for contradiction that for some positive integer k ,

$$(\iota_d)_\# \mu_q = \mathfrak{f}_\# \mathcal{L}(X),$$

where X has a bounded Lebesgue density in $\mathbb{R}^k$ and $\mathfrak{f}(x) = b + Ax$ has full-column-rank A . Equality of laws gives equality of topological supports. By Lemma A.8, the left support has affine dimension three. By Proposition A.13, the right support has affine dimension k . Thus

$$k = 3.$$

Every positive latent dimension $k \neq 3$, including $k = 1, 2$ and every $k > 3$ compatible with full column rank, is excluded by dimension alone.

For $k = 3$, let $L = b + \text{im}(A)$. Proposition A.13 gives

$$\mathfrak{f}_{\#}\mathcal{L}(X) \ll \mathcal{H}_L^3, \quad L = \text{aff}(\text{supp}(\mathfrak{f}_{\#}\mathcal{L}(X))).$$

Equality of laws and Lemma A.8 identify L with $\text{aff}(\iota_d(S_q))$ and give

$$\mathcal{H}_L^3(\iota_d(S_q)) = 0.$$

Absolute continuity implies

$$(\mathfrak{f}_{\#}\mathcal{L}(X))(\iota_d(S_q)) = 0,$$

whereas exact support and the embedding give

$$((\iota_d)_{\#}\mu_q)(\iota_d(S_q)) = \mu_q(S_q) = 1,$$

a contradiction. This is the missing dimension-three argument that curvature alone would not supply.

If a zero-dimensional latent variable is admitted, its affine image is a single point and has affine-support dimension zero, again contradicting Lemma A.8. If $k > d + 1$, no $(d + 1) \times k$ matrix has full column rank. Thus every nonnegative latent dimension is covered. In the minimal case $d = 3$, the branch sheet spans all three lower-coefficient coordinates, its monic embedding lies in $\mathbb{R}^4$, and the same $k = 3$ null-volume contradiction applies.

The conclusion does not address rank-deficient affine maps paired with singular latent laws or any broader comparison class. It uses neither a no-forced-root condition nor any root-section estimate or root bound from the cited source. Propositions A.11 and A.12, together with the exact monic embedding and affine-pushforward density calculation, provide both the dimension and within-hull-volume arguments needed for this precisely limited nonmembership result. $\square$

A.7 The Sufficient Theorem and Baseline Reduction

Proposition A.15 (Complete averaged-intercept sufficient conclusion). *Fix the basic setting. The derivative envelope satisfies*

$$L_{d,R,\Theta} \leq dB^{d-1} + R \sum_{k=1}^{d-1} kB^{k-1} < \infty,$$

with the sum zero when $d = 1$. Under Assumption 1, simultaneously for every $\mu \in \mathcal{D}$ and $I \in \mathcal{J}(\Theta)$,

$$\Pr_{\mu}(Z_{\alpha} \cap I \neq \emptyset) \leq L_{d,R,\Theta}|I| \int K_{\mu}(\beta) \pi_{\mu}(d\beta) \leq \bar{\kappa}_{\mathcal{D}} L_{d,R,\Theta}|I|,$$

and

$$C_{\mathcal{D}} \leq \bar{\kappa}_{\mathcal{D}} L_{d,R,\Theta} < \infty.$$

Separately, under Assumption 2, every $\mu \in \mathcal{D}_{\text{RI}}$ is cube-supported and, for π_{μ} -almost every β ,

$$f_{\mu}(a_0 \mid \beta) = \frac{\mathbf{1}\{|a_0 - G_{\mu}(\beta)| \leq \rho_{\mu}(\beta)\}}{2\rho_{\mu}(\beta)}, \quad K_{\mu}(\beta) = \frac{1}{2\rho_{\mu}(\beta)}.$$

For every such law and every $I \in \mathcal{J}(\Theta)$,

$$\bar{\kappa}_{\mathcal{D}_{\text{RI}}} = \frac{1}{2}M_{\text{RI}}, \quad \Pr_{\mu}(Z_{\alpha} \cap I \neq \emptyset) \leq \frac{L_{d,R,\Theta}|I|}{2} \int \rho_{\mu}^{-1} d\pi_{\mu},$$

and

$$C_{\mathcal{D}_{\text{RI}}} \leq \frac{1}{2} L_{d,R,\Theta} M_{\text{RI}} < \infty.$$

If $\rho_\mu = r_\mu > 0$ almost surely, then

$$K_\mu = \frac{1}{2r_\mu} \quad \pi_\mu\text{-almost surely}, \quad \Pr_\mu(Z_\alpha \cap I \neq \emptyset) \leq \frac{L_{d,R,\Theta}}{2r_\mu} |I|.$$

For every nonempty class $\mathcal{F}$ of such fixed-width members obeying the support condition and $\sup_{\mu \in \mathcal{F}} r_\mu^{-1} < \infty$,

$$\bar{\kappa}_{\mathcal{F}} = \frac{1}{2} \sup_{\mu \in \mathcal{F}} r_\mu^{-1}, \quad C_{\mathcal{F}} \leq \frac{L_{d,R,\Theta}}{2} \sup_{\mu \in \mathcal{F}} r_\mu^{-1} < \infty.$$

For every $d \geq 3$, $R > 0$, and $q \in (0, 1)$,

$$\text{supp}(\mu_q) = \{(a_0, z, z^2/R, 0, \dots, 0) : |z| \leq R/2, |a_0| \leq \rho_q(z)\} \subseteq [-R, R]^d, \quad \Pr(\rho_q(Z) > 0) = 1.$$

For $a_0 \in \mathbb{R}$ and $0 < |z| \leq R/2$, one conditional-density version satisfies

$$f_{\mu_q}(a_0 \mid \beta_q(z)) = \frac{\mathbf{1}\{|a_0| \leq \rho_q(z)\}}{2\rho_q(z)}, \quad K_{\mu_q}(\beta_q(z)) = \frac{1}{R(2|z|/R)^q}.$$

Moreover,

$$\text{ess sup}_{\beta \sim \pi_{\mu_q}} K_{\mu_q}(\beta) = \infty, \quad \int K_{\mu_q} d\pi_{\mu_q} = \frac{1}{R(1-q)},$$

and, for every $I \in \mathcal{I}(\Theta)$,

$$\Pr_{\mu_q}(Z_\alpha \cap I \neq \emptyset) \leq \frac{L_{d,R,\Theta}|I|}{R(1-q)}, \quad C_{\{\mu_q\}} \leq \frac{L_{d,R,\Theta}}{R(1-q)}.$$

The support has affine-hull dimension three and zero induced three-dimensional Hausdorff volume there, and μ_q is not a full-column-rank affine pushforward of a bounded-support, bounded-Lebesgue-density latent law under the fixed-monic convention. This last conclusion does not extend by assertion to rank-deficient maps or singular latent laws.

All displayed conclusions are one-way and sufficient. They assert no necessity, converse, indexed uniform polynomial dependence on (d, R) , root simplicity, root separation, or transversality, and supply neither a learner nor an online guarantee.

Proof. Lemma A.1 gives the first display, including the empty-sum case. Proposition A.3 gives the general per-law inequality and closes both defining suprema for $C_{\mathcal{D}}$, without any mode conversion.

Under the separately scoped random-intercept assumption, Proposition A.4 proves cube support, the conditional density, exact cap, per-law cap integral, and exact class envelope. It thereby verifies Assumption 1 for $\mathcal{D}_{\text{RI}}$ rather than assuming it. Proposition A.5 gives the per-law and class root bounds, and Proposition A.6 gives the fixed-width statements under precisely the class-uniform reciprocal-width condition. None of these results needs a density or coordinate independence assumption for π_μ .

For the explicit witness, Proposition A.7 gives the support equality, Proposition A.8 gives cube containment and almost-sure positivity, and Proposition A.9 gives the selected conditional density and exact nonzero-fiber cap. The collapsed zero fiber remains in topological support but has zero conditioning-marginal mass and is assigned a genuine fallback density. Lemma A.6 gives the infinite essential supremum, Lemma A.7 gives the exact finite integral, and Proposition A.10 gives the interval and singleton bounds.

Propositions A.11 and A.12 give the two independent structural facts. Proposition A.14 applies them only to the full-column-rank latent-Lebesgue-density convention, so the conclusion does not rely on curvature alone and does not overstate the excluded class.

The final scope paragraph records the direction of the established implications. None of the cited propositions supplies a reverse implication, a polynomial-family bound, or a root-regularity conclusion. Thus every displayed rate retains its exact exposed parameters and no unstated inference is added. $\square$

Lemma A.9 (A bounded joint density implies the averaged intercept cap). *Fix the basic cube setting and let $\mathcal{A}$ be a nonempty class of Borel probability laws on $[-R, R]^d$. Suppose there is a common $\kappa_{\text{joint}} \in (0, \infty)$ such that every $\nu \in \mathcal{A}$ has a Borel density $h_\nu(a_0, \beta)$ with respect to $\mathcal{L}^d$, extended by zero outside the cube, and*

$$0 \leq h_\nu(a_0, \beta) \leq \kappa_{\text{joint}} \quad \text{for Lebesgue-almost every } (a_0, \beta).$$

Then $\mathcal{A}$ satisfies Assumption 1, and one may choose measurable conditional-density and cap versions such that

$$\int K_\nu(\beta) \pi_\nu(d\beta) \leq \kappa_{\text{joint}} (2R)^{d-1} \quad \text{for every } \nu \in \mathcal{A}.$$

Consequently,

$$\bar{\kappa}_{\mathcal{A}} \leq \kappa_{\text{joint}} (2R)^{d-1}.$$

For $d = 1$, $(2R)^0 = 1$ and the same statements hold.

Proof. First suppose $d \geq 2$ and fix $\nu \in \mathcal{A}$. Define the higher-coordinate marginal density

$$g_\nu(\beta) := \int_{\mathbb{R}} h_\nu(a_0, \beta) da_0.$$

Tonelli's theorem makes g_ν Borel and gives $\pi_\nu(d\beta) = g_\nu(\beta)d\beta$. Because h_ν is supported on the cube and bounded by κ_{joint} ,

$$0 \leq g_\nu(\beta) \leq 2R\kappa_{\text{joint}} \quad \text{for Lebesgue-almost every } \beta, \quad \int g_\nu(\beta)d\beta = 1.$$

Thus the measurable set

$$G_\nu := \{\beta : 0 < g_\nu(\beta) < \infty\}$$

has full π_ν -measure: $\{g_\nu = 0\}$ has zero marginal mass, and $\{g_\nu = \infty\}$ is Lebesgue-null by integrability and also has zero marginal mass.

Define the sectionwise essential supremum

$$H_\nu(\beta) := \text{ess sup}_{a_0 \in \mathbb{R}} h_\nu(a_0, \beta).$$

It has a measurable extended-valued version. Indeed, for every rational $t \geq 0$, Tonelli makes

$$m_t(\beta) := \int_{-R}^R \mathbf{1}\{h_\nu(a_0, \beta) > t\} da_0$$

measurable, and $\{H_\nu > t\} = \{m_t > 0\}$. The rational superlevel sets determine a measurable version of H_ν . Fubini applied to the joint bound gives

$$H_\nu(\beta) \leq \kappa_{\text{joint}} \quad \text{for Lebesgue-almost every } \beta.$$

Now define

$$f_\nu(a_0 \mid \beta) = \begin{cases} h_\nu(a_0, \beta)/g_\nu(\beta), & \beta \in G_\nu, \\ \mathbf{1}\{|a_0| \leq R\}/(2R), & \beta \notin G_\nu. \end{cases}$$

This is jointly measurable and makes no division on a zero or infinite marginal-density fiber. Each line integrates to one: on G_ν this is the definition of g_ν , and on G_ν^c it follows from $R > 0$. For Borel sets $A \subseteq \mathbb{R}$ and $D \subseteq \mathbb{R}^{d-1}$, Tonelli gives

$$\begin{aligned} \int_D \int_A f_\nu(a_0 \mid \beta) da_0 \pi_\nu(d\beta) &= \int_{D \cap G_\nu} \int_A h_\nu(a_0, \beta) da_0 d\beta \\ &= \nu\{\alpha_0 \in A, \beta \in D\}. \end{aligned}$$

The omitted part $D \cap G_\nu^c$ has zero marginal and joint-law mass. Thus f_ν is a jointly measurable regular conditional density of the actual intercept given the actual higher coordinates, extended by zero outside $[-R, R]$.

Its measurable essential-supremum version is

$$K_\nu(\beta) = \begin{cases} H_\nu(\beta)/g_\nu(\beta), & \beta \in G_\nu, \\ 1/(2R), & \beta \notin G_\nu. \end{cases}$$

On G_ν , multiplication by the finite positive number $g_\nu(\beta)$ commutes with the Lebesgue essential supremum, and hence, for Lebesgue-almost every $\beta \in G_\nu$,

$$g_\nu(\beta)K_\nu(\beta) = H_\nu(\beta) \leq \kappa_{\text{joint}}.$$

The complement of G_ν is marginal-null, so its fallback cap contributes nothing. Integration over the exact higher-coordinate cube gives

$$\begin{aligned} \int K_\nu(\beta) \pi_\nu(d\beta) &= \int_{G_\nu \cap [-R, R]^{d-1}} g_\nu(\beta)K_\nu(\beta) d\beta \\ &\leq \kappa_{\text{joint}} \text{Leb}^{d-1}([-R, R]^{d-1}) \\ &= \kappa_{\text{joint}}(2R)^{d-1}. \end{aligned}$$

This establishes joint measurability of the conditional density, measurability of its cap, and finite cap integral for $d \geq 2$.

When $d = 1$, the conditioning space is the one-point empty-tuple space. The conditional density is the one-dimensional joint density $h_\nu(a_0)$ and

$$K_\nu = \text{ess sup}_{a_0} h_\nu(a_0) \leq \kappa_{\text{joint}} = \kappa_{\text{joint}}(2R)^0.$$

The cap integral is evaluation at that one point. Taking the class supremum proves the asserted envelope bound. The density and cap constructions verify every part of Assumption 1 for $\mathcal{A}$; the joint-density premise is confined to this comparison. $\square$

Proposition A.16 (Recovery of the monic bounded-joint-density baseline). *Under the basic setting and the hypotheses of Lemma A.9, that lemma and Proposition A.3 imply that every $\nu \in \mathcal{A}$ and $I \in \mathcal{I}(\Theta)$ satisfy*

$$\Pr_\nu(Z_\alpha \cap I \neq \emptyset) \leq \kappa_{\text{joint}}(2R)^{d-1} L_{d,R,\Theta}|I|,$$

and

$$C_{\mathcal{A}} \leq \kappa_{\text{joint}}(2R)^{d-1} L_{d,R,\Theta}.$$

If $B = 1$, then

$$C_{\mathcal{A}} \leq \kappa_{\text{joint}}(2R)^{d-1} \left(d + \frac{Rd(d-1)}{2} \right).$$

For $d = 1$, $(2R)^0 = 1$ and the bracket equals one.

Proof. Lemma A.9 constructs the required conditional density and cap versions for $\mathcal{A}$, verifies Assumption 1, and proves

$$\bar{\kappa}_{\mathcal{A}} \leq \kappa_{\text{joint}}(2R)^{d-1}.$$

Proposition A.3 therefore applies to the comparison class. Its per-law and class conclusions yield the first two displays without using the joint-density premise anywhere in Proposition A.15.

Lemma A.1 gives

$$L_{d,R,\Theta} \leq dB^{d-1} + R \sum_{k=1}^{d-1} kB^{k-1}.$$

When $B = 1$, every power equals one and

$$\sum_{k=1}^{d-1} k = \frac{d(d-1)}{2},$$

with both sides zero for $d = 1$. Substitution gives the final display. Every factor is exact: conditioning on the intercept leaves the higher-coordinate cube volume $(2R)^{d-1}$, and the monic derivative contributes $d + Rd(d-1)/2$. Thus the $B = 1$ statement recovers the monic ambient-volume scale rather than an asymptotic surrogate. In the absence of the optional joint-density premise, the general averaged-envelope theorem and both constructive specializations remain unchanged. $\square$

A.8 Proof of the Main Theorem

Proof of Theorem 3.1. Lemma A.1 proves the displayed deterministic bound on $L_{d,R,\Theta}$. Under Assumption 1, Proposition A.3 proves the exact general per-law, per-interval inequality and the bound on $C_{\mathcal{D}}$.

Under the separately scoped Assumption 2, Proposition A.4 proves cube support, the jointly measurable conditional density, the exact reciprocal-width cap, and the averaged envelope. Proposition A.5 proves the corresponding root-hitting bounds, and Proposition A.6 proves every fixed-width statement.

For the explicit witness, Propositions A.7 and A.8 prove exact support, cube membership, and almost-sure positivity. Lemma A.5 and Proposition A.9 prove the actual conditioning identity, the measurable fallback construction, and the exact nonzero-fiber cap. Lemmas A.6 and A.7 prove, respectively, infinite essential cap and the exact finite cap integral, while Proposition A.10 proves the singleton interval and $C_{\{\mu_q\}}$ bounds.

Propositions A.11 and A.12 prove the exact affine dimension and within-hull nullity. Lemma A.8 preserves those properties in descending fixed-monic coordinates, and Propositions A.13 and A.14 prove the precisely limited full-column-rank affine-latent exclusion. Proposition A.15 collects these conclusions with their original quantifiers and records the sufficient-only boundary. Every displayed dependence in Theorem 3.1 is therefore proved with no hidden constant, probability-mode conversion, or horizon change. $\square$

Proof of Corollary 3.1. Lemma A.9 constructs the conditional-density versions and proves $\bar{\kappa}_{\mathcal{A}} \leq \kappa_{\text{joint}}(2R)^{d-1}$ under the corollary's optional premise. Proposition A.16 then gives the interval and class bounds and evaluates the derivative sum exactly when $B = 1$, including the $d = 1$ empty-tuple case. $\square$